

Agentic Pilot: An Evidence-Driven Local Autonomous AI Agent Framework for Privacy-Preserving Intelligent Task Automation

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Abstract—LLM-based assistants that operate software for a user often send screenshots and page content to hosted models, often keep no memory across sessions, and can declare a task finished when it is not. We present Agentic Pilot, an open-source agent that runs its language and vision models locally through Ollama and drives Chromium through Playwright. The agent is a LangGraph state machine with separate observation, planning, execution, verification and recovery nodes, and it may report a task complete only when a completion predicate evaluated on the live page holds; every evaluation is stored with before/after screenshots and a page snapshot. Grounding uses a compact list of Document Object Model (DOM) elements and calls a local vision model only when the planner cannot act on it. Calls are routed to small models by role, CAPTCHAs halt the task for the user, and high-risk intents need human approval. We formalize the gate and define a false completion rate. On 35 constructed browser end states, the gate withheld completion in 15 of the 22 states that did not meet their goal, all of which a dispatch-only rule accepts, and rejected 1 of 13 states that did, at a median 33.8 ms per decision; each miss points to a specific missing check. A code-level audit reports which mechanisms are active on the live path. We also design an extension to files, processes and native applications with accessibility-tree grounding, typed postconditions and deterministic permission tiers, and give an end-to-end evaluation protocol for local hardware.

Index Terms—autonomous agents, large language models, local inference, privacy-preserving computing, task automation, execution verification, human-in-the-loop

I. INTRODUCTION

Assistants built on large language models (LLMs) increasingly act on a user’s behalf: they open pages, fill forms and read results. Several of the web and GUI agents reviewed in Section II send each observation, often a full screenshot, to a hosted multimodal model [1]–[4]. For a personal assistant this is a poor fit. The screen of a logged-in user carries names, messages and account data, and leakage of personal information from language models has been measured even after scrubbing [5]. Few of these systems keep memory across sessions, and most operate either in a browser or in one other environment rather than across the applications a person uses (Table I).

A second problem is independent of where the model runs. An agent can believe it has finished when it has not. We call this *premature completion*: the agent emits a terminal “done” while the environment does not satisfy the user’s goal. A typical case is a search request in which the agent

opens the search engine, sees the page load without error, and stops before any query is issued. Every individual browser call succeeded; the task did not. Benchmarks that check final environment state show how wide this gap is [6], [7], and self-assessment by the model is not a reliable remedy, since intrinsic self-correction without external feedback does not improve, and can degrade, reasoning accuracy [8].

This paper describes Agentic Pilot, an open-source agent that runs its language and vision models locally through Ollama and drives Chromium through Playwright. Its central rule is that a task may be reported complete only when a check on the observed environment passes, with the inputs to that check written to disk as evidence. Observation, planning, execution, verification and recovery are separate nodes of an explicit LangGraph state machine, so every transition can be inspected and logged.

We make the following contributions:

- 1) An evidence-gated execution model that separates action dispatch from task completion, formalized as a completion predicate over observations, together with a *false completion rate* (FCR) metric for measuring premature completion (Section IV-C).
- 2) The architecture and implementation of a local agent that combines this gate with role-based routing across local models, grounding in the page’s Document Object Model (DOM) first with a vision-model fallback, CAPTCHA detection that halts rather than bypasses, a task-level human approval gate, and persistent per-step evidence (Sections III–V).
- 3) A controlled, reproducible study of the completion gate on 35 constructed browser end states, which quantifies both the premature completions it prevents and the ones it misses (Section VII).
- 4) A design for extending the framework from the browser to the local system (files, processes and native applications) with accessibility-tree-first grounding, typed desktop verification predicates and deterministic permission tiers; we state precisely which parts of it exist in code today (Section IV-G).
- 5) A code-level audit of which mechanisms are active on the live execution path, and an end-to-end evaluation protocol with ablations to be run on local hardware (Sections V and VI).

Sections II–IV cover related work, architecture and method; Sections V–VII the implementation, setup and results; Sections VIII–X security, limitations and conclusions.

II. RELATED WORK

A. Reasoning-and-acting agents

ReAct interleaves free-text reasoning with tool actions and observations [9], building on chain-of-thought prompting [10]. Toolformer shows that a model can learn when to call an external API [11]. Reflexion stores verbal self-critiques in an episodic buffer and retries with them [12]; RCI applies recursive self-criticism to computer tasks [13]; Voyager accumulates a skill library and uses a model-based self-check before committing a skill [14]. In all of these, the decision that a task is finished is made by the model, sometimes with a critic that is also a model. Huang et al. find that such intrinsic self-correction does not reliably improve reasoning without external feedback [8]. Agentic Pilot moves the completion decision out of the model and into a check on the observed page.

B. Web agents and benchmarks

Early web agents were trained with reinforcement or imitation learning on MiniWoB-style pages [15]. Mind2Web pairs a small ranking model that filters candidate DOM elements with an LLM that chooses the action [16], and WebArena evaluates agents on self-hosted sites with functional correctness checks [6]. WebVoyager drives live sites from screenshots and uses a multimodal model to judge saved trajectories after the episode [1]. The correctness checks in these works belong to the benchmark and run after the episode ends. Our gate runs inside the agent at the moment it tries to stop.

C. Desktop, OS and mobile GUI agents

OSWorld provides execution-based checkers for tasks in real operating systems and reports a large gap between agents and people [7]. UFO controls Windows applications through UI Automation and asks the user to confirm sensitive actions [3]. AppAgent learns reusable documentation of smartphone apps [4], and GPTDroid and QTypist apply LLMs to mobile GUI testing and text-input generation [17], [18]. These systems either rely on hosted models or target testing rather than verified completion of user tasks. Our desktop extension (Section IV-G) borrows UFO’s accessibility-first control and OSWorld’s state-based checking, but places the checks inside the agent loop.

D. Visual grounding for GUI interaction

SeeAct finds that grounding a chosen action onto the right element is the main bottleneck for multimodal web agents, and that combining HTML with the image outperforms Set-of-Mark overlays [2], [19]. CogAgent and SeeClick train vision-language models that locate elements from screenshots alone [20], [21], and OmniParser converts screenshots into structured element lists [22]. These results motivate our ordering: structured DOM grounding first, pixel-level grounding only when the DOM route fails.

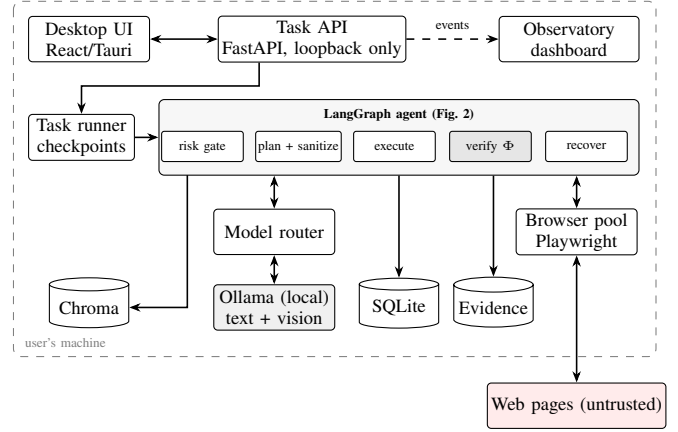


Fig. 1. Components of Agentic Pilot. Everything inside the dashed boundary runs on the user’s machine, including model inference. Web pages are the main untrusted input, and browsing is the main outbound traffic. The Observatory also reads the task database and evidence files.

E. Agent memory and retrieval

Retrieval-augmented generation couples a model with a non-parametric store [23], usually indexed with sentence embeddings [24]. Generative Agents score memories by recency, importance and relevance [25], and MemGPT pages context between tiers in the manner of an operating system [26]. None of these ties a stored memory to verified evidence of the outcome it describes, which is what an execution agent needs if it is to learn from its own runs.

F. Security of LLM agents

Indirect prompt injection places instructions in content the agent reads, such as a web page [27]. Liu et al. formalize such attacks and benchmark defenses [28]; InjecAgent and AgentDojo measure how often tool-using agents follow injected instructions [29], [30]. A browser agent reads untrusted text at every step, so it needs both input filtering and a policy that keeps high-impact actions under human control.

G. Local and on-device inference

Open-weight families such as Qwen2.5 and Qwen2-VL provide text and vision models at sizes that run on consumer hardware [31], [32], and weight quantization further reduces their memory footprint [33]. Surveys of LLM agents organize the design space into profile, memory, planning and action components and list trustworthiness among the open problems [34], [35]. How often models of this size finish GUI tasks whose outcome is independently checked has received little measurement, and our evaluation protocol targets that question.

Table I summarizes the comparison. Cells record only what each paper reports.

III. SYSTEM ARCHITECTURE

Fig. 1 shows the components and the trust boundary. A desktop client submits a natural-language task to a FastAPI service bound to the loopback interface. The task runner

TABLE I

CAPABILITIES REPORTED BY REPRESENTATIVE SYSTEMS. LOC: RUNS ON OPEN OR LOCALLY HOSTABLE MODELS; WEB/DESK: TARGET ENVIRONMENT; GATE: COMPLETION GATED AT RUN TIME BY AN IN-AGENT ENVIRONMENT CHECK; EVID: PERSISTS STATE ARTIFACTS; MEM: CROSS-STEP OR CROSS-EPISEODE MEMORY; INJ: PROMPT-INJECTION DEFENSE; APPR: HUMAN APPROVAL FOR RISKY ACTIONS. **Y** REPORTED, **P** PARTIAL OR WEAKER FORM, **N** NOT SUPPORTED, – NOT REPORTED. THE LAST ROW REFLECTS THE AUDITED CODE (SECTION V).

System	Loc	Web	Desk	Gate	Evid	Mem	Inj	Appr
ReAct [9]	N	P	N	N	N	N	N	N
Reflexion [12]	–	P	N	P	N	Y	N	N
WebArena [6]	–	Y	N	P	–	N	N	N
Mind2Web [16]	Y	Y	N	N	P	N	N	N
WebVoyager [1]	N	Y	N	P	Y	N	N	N
SeeAct [2]	P	Y	N	N	–	N	N	P
OSWorld [7]	Y	Y	Y	P	–	N	N	N
UFO [3]	N	P	Y	–	–	P	N	Y
AppAgent [4]	N	N	N	–	–	Y	N	N
CogAgent [20]	Y	Y	–	N	N	N	N	N
MemGPT [26]	–	N	N	N	N	Y	N	N
AgentDojo [30]	Y	N	N	P	–	N	Y	–
Agentic Pilot	Y	Y	P ^a	Y ^b	Y	P ^c	P ^d	P ^e

^aDesigned (Section IV-G); a prototype executor exists but is not wired into the agent.

^bRule-based predicate Φ ; see its measured blind spots in Section VII. ^cOutcomes are written; retrieval is not active on the live path. ^dPattern denylist on element text.

^eTask-level, triggered by the intent’s action type or model-assigned risk level.

AppAgent targets Android, which we do not count as desktop.

queues it, executes the agent graph, checkpoints the state after every node so that a paused or interrupted task can resume, and cancels tasks that exceed a wall-clock limit (5 min by default).

The agent is a compiled LangGraph state machine with eleven nodes (Fig. 2). A typed state record carries the parsed intent, the plan and step index, the page manifest, the planned action, the action history, retry and model-call counters, routing decisions and the status; nodes return partial updates and the routers read only the status and a few flags. The single terminal node, `complete`, persists the outcome, writes memory records and decides whether the browser stays open for the user. Model calls go through a router (Section IV-E) and a gateway that requests JSON output from a local Ollama server, validates it against Pydantic schemas and re-prompts on malformed output. A pool of Playwright contexts over one headless Chromium gives each task its own page. Task events go to SQLite and to a broadcaster; the Observatory, a separate process, subscribes to that stream and reads the database and evidence files to display plans, routing decisions, verification outcomes and screenshots. It only issues read queries and has no command channel to the agent.

Two inputs cross into the agent: the user’s task text, which is trusted, and page content, which is not. Page content reaches the models as element text, page title and URL, and, on the vision path, screenshots. Apart from one-time downloads of model weights (the Ollama models and, on first use, the default embedding model of the vector store), outbound traffic in the default configuration consists of the browser’s requests to the sites a task visits; model traffic stays on the local host unless the user points the Ollama address elsewhere.

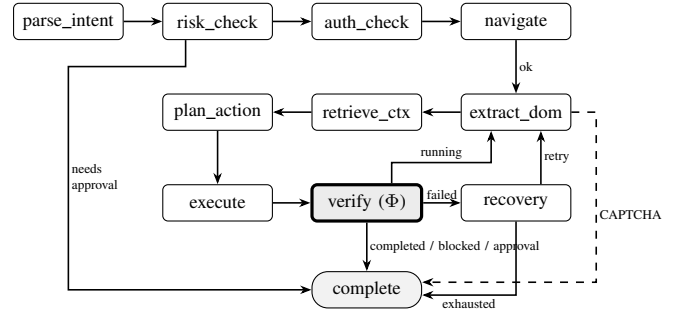


Fig. 2. Execution graph compiled in `graph.py`. Only the shaded `verify` node can end a task as `COMPLETED`, and only when Φ holds. A CAPTCHA seen during observation ends it as `BLOCKED`. Not drawn: failures in `parse_intent` and `navigate` also route to `recovery`.

IV. METHODOLOGY

A. Task Decomposition and Planning

The `parse_intent` node asks a lightweight model for a structured intent: action, target site, target and content strings, risk level and confidence. If the request contains a connective (“and”, “then”, a comma) or more than eight words, a planning model decomposes it into steps $P = (g_1, \dots, g_n)$, each with a description, an action type and an expected outcome; a request to enter literal text becomes a single text-entry step, and anything else a one-step plan.

Execution then chooses one action per iteration. `plan_action` gives the model the goal, the plan with the current step marked, the page title, URL and state, and the element manifest, and asks for one of `navigate`, `click`, `type_text`, `press_key`, `scroll`, `extract`, `complete` or `need_help`. Deterministic rules then adjust the choice: exact-text requests are grounded to the best-scoring visible text field, repeated navigation to the current site is suppressed, and plan steps that name clicking a result, extracting or reporting are mapped to those actions. Some rules are site-specific; on Google with a search goal the agent types the query and presses Enter rather than loading a results URL directly (see Section VIII).

B. Grounding: DOM First, Vision on Demand

Each observation starts with a DOM pass over buttons, inputs, text areas, selects, links, elements with button or link roles, and editable regions. For each element the extractor records tag, role, accessible label, up to 120 characters of text, placeholder, type, visibility, enabled state, bounding box, and three selectors (an identifier attribute it writes into the live DOM, a CSS selector and an XPath, tried in that order). Only interactable elements are kept; they are ranked with submit-like controls first, deduplicated, limited to five links, stripped of cookie and advertising labels, and capped at 50.

The vision model is used only when the planner returns `need_help`, by choice or because its call failed. The agent sends a screenshot and receives an action with a target point given as fractions of the 1280×800 viewport, where it clicks

and, if needed, types. Such actions are not checked individually; a wrong click is caught, if at all, by the completion predicate. Identical screenshot-goal pairs are served from a cache.

C. Evidence-Gated Verification Model

We model an episode as a sequence of observations $o_t = \Omega(s_t)$, where s_t is the browser state after step t and Ω returns the page URL, the page title, the interactive-element manifest and a screenshot. The planner maps $(g, o_t, \mathcal{M})$, with g the user goal and $\mathcal{M}$ the retrieved context, to an action a_t . The executor applies a_t and returns a dispatch flag $\alpha_t \in \{0, 1\}$ that is 1 when the browser call returned without an exception or a failed low-level check (e.g., reading back the value of a text field).

a) *Blocking predicate*: $\beta(o) = 1$ when the observation shows a state in which further autonomous action is either impossible or not permitted: a `chrome-error://URL`, an HTTP error title (404, 500, 502, 503), or a CAPTCHA/bot-check page. A CAPTCHA, detected by a separate check that runs before Φ , forces the terminal state BLOCKED; the error cases make Φ false, so the episode continues.

b) *Step predicates*: The decomposer produces a plan $P = (g_1, \dots, g_n)$. In the current implementation the per-step predicate is weak: $\phi_i(o_{t+1}) := \alpha_t$, i.e., a step is marked complete when its action dispatched successfully, and the plan index advances.

c) *Task predicate*: Completion is decided by a separate predicate $\Phi(g, o, \hat{y}, P)$ evaluated only when the planner emits `complete`, when the plan index reaches n , or when a requested text entry has been dispatched. With $\hat{y}$ the extracted answer,

$$\Phi = \neg\beta(o) \wedge \neg H(g, o) \wedge T(g, o) \wedge X(g, \hat{y}) \wedge R(P, \hat{y}), \quad (1)$$

where H is true when the goal requires search or extraction but o is still a search-engine home page; T requires that any literal text requested in g appears in an input, textarea or contenteditable element of the live DOM; X requires, for extraction goals, a non-empty structured extraction or an answer longer than 40 characters that is not a restated navigation instruction; and R requires that no plan step remains pending unless an answer has already been produced. Each conjunct is optional and is only instantiated when g triggers it.

d) *Completion rule*: The agent reports success iff $\Phi = 1$, and every evaluation of Φ is written to the evidence store together with its expected and observed values. A rejected evaluation does not terminate the episode; control returns to observation and planning until a budget of $K = 15$ model calls (intent parsing, decomposition and planning) is exhausted. A final gate re-checks that navigation actually occurred when the intent named a target site, and downgrades the outcome to FAILED otherwise.

e) *Property*: Dispatch success does not imply task success:

$$\alpha_t = 1 \not\Rightarrow \Phi(g, o_{t+1}, \hat{y}, P) = 1. \quad (2)$$

A successful `navigate` to a search engine home page is the canonical counterexample: $\alpha_t = 1$ but $H = 1$, hence $\Phi = 0$.

f) *False completion rate*: Let $\mathcal{T}_c$ be the set of episodes the agent reports as completed, and let Φ^* be an independent, task-specific oracle predicate written by the experimenter (not by the agent). Then

$$\text{FCR} = \frac{|\{\tau \in \mathcal{T}_c : \Phi^*(\tau) = 0\}|}{|\mathcal{T}_c|}. \quad (3)$$

FCR is defined over reported completions, so it measures how often the agent claims success it has not achieved; task success rate is reported separately.

D. Failure Classification and Recovery

A failed action is first retried through a fresh observation by the `verify` node, up to three times. Later failures go to the recovery node, which matches the error text against patterns for seven types (`captcha`, `transient`, `element_not_found`, `verification_failed`, `navigation_failed`, `vision_needed`, `llm_failure`) or `unknown`. A CAPTCHA ends the task as BLOCKED. Other failures are assigned a strategy from the ladder `retry` $\rightarrow$ `alternative_selector` $\rightarrow$ `vision_fallback` $\rightarrow$ `replan`, one level per two attempts, with type-specific shortcuts; after six attempts the task fails. Each decision is stored as a recovery record. In the current code the chosen strategy is recorded but not executed: the strategy field is not declared in the state schema and is dropped, so every non-terminal recovery re-observes and re-plans.

E. Role-Based Local Model Routing

A deterministic analyzer maps each call's type, text and flags to required capabilities and a role: recovery calls to a recovery model; image inputs or screen-related wording to a vision model; intent parsing and simple navigation to a lightweight model; decomposition and multi-clause requests to a planner; the rest to a general model. Each role has a configured, ordered candidate list; the defaults are small Qwen2.5, Qwen3.5 and DeepSeek-R1 models of 1.5B–7B parameters for text roles and `qwen3-vl:2b` or `moondream` for vision. The first candidate the local Ollama server reports as installed is used (the list is refreshed every 300 s). To limit model reloads, the router keeps the active model whenever it has the required capabilities, except for recovery calls; with no installed candidate it falls back to a fallback and then a base model. A single-model mode serves as an ablation. A switch cool-down is configurable but not read by the router.

F. Episodic Memory

On termination, `complete` writes the outcome to SQLite and to a Chroma collection: for a success, the goal, final URL and number of model calls; for a failure, the error and URL. The retrieval side exists: a context router queries memory (and, for longer or procedural tasks, a knowledge collection), and a builder deduplicates the results and fits them into a 3,500-token

budget inside an untrusted-content banner in the planning prompt. On the live path, however, the runner initializes the retrieved lists as empty rather than unset, which the router treats as already retrieved; memory is written but not read. Since the stored records hold no selectors or action sequences, repairing the read path is necessary but not sufficient for learning from past runs.

G. Extension to Local System Access

The browser loop carries over to the desktop if the accessibility tree replaces the DOM, a window screenshot replaces the page screenshot, and typed operating-system predicates replace URL and DOM checks. Today the repository contains a desktop executor wrapping PyAutoGUI (click, type, hotkey, screenshot) and psutil (process listing); no node, action type or API route calls it, its only test checks that its methods exist, and without a display it silently returns a fixed screen size and empty screenshots. A file helper reads and writes text files but is not called by the agent and does not confine paths to its workspace. The rest of this subsection is a design (PROPOSED).

a) Actions and grounding: The action space covers files (list, hash, read, create, overwrite with backup, move, trash, delete), processes (list, launch an allow-listed program, terminate), applications (open, focus, close) and UI controls (invoke, set value, type, hotkey). Grounding reads the foreground window’s accessibility tree (Windows UI Automation, macOS Accessibility, AT-SPI) into a manifest shaped like the DOM manifest, with names sanitized. Vision on a window screenshot is used when the tree exposes too few actionable elements or the planner asks for help, and a coordinate action counts as done only after its postcondition holds.

b) Verification: Every mutating action carries typed, exact postconditions evaluated after execution and stored with before/after values (Table II). File predicates compare SHA-256 digests, sizes and modification times; process predicates identify a process by PID and creation time, which guards against PID reuse; window and accessibility predicates compare titles, focus, values and states. Read-only actions must leave a manifest of the workspace unchanged.

c) Permission tiers and enforcement: Tiers are computed from the concrete action and arguments, never from the model’s risk label. T0 (observe) runs freely. T1 (reversible: the inverse is recorded, e.g., a backed-up overwrite or a move to trash) runs automatically only if the user enabled automatic approval of low-risk actions. T2 (destructive: permanent deletion, terminating a process the agent did not start, invoking controls labelled Delete, Discard, Send or Pay) and T3 (network-capable programs, password fields) always need approval; secrets are injected by the executor and never pass through the model. Paths outside allow-listed roots, credential directories, inline-code interpreters and privilege elevation are refused. A new `action_gate` node between planning and execution applies the tier, a path jail that resolves symbolic links before the containment test, and a dry-run preview; approvals are bound to a hash of the exact action so arguments

TABLE II
PROPOSED DESKTOP POSTCONDITIONS AND DEFAULT PERMISSION TIERS (DESIGN, NOT IMPLEMENTED). h_0, t_0 : VALUES CAPTURED BEFORE THE ACTION; (pid, t_c) : PROCESS ID AND CREATION TIME.

Action	Postcondition checked after execution	Tier
file read, list, hash	workspace manifest unchanged	T0
create file	file exists $\wedge$ sha256 = sha256(content)	T1
overwrite	backup of pre-image stored $\wedge$ hash equals new content	T1
move	source absent $\wedge$ sha256(dst) = $h_0(src)$	T1
move to trash	path absent $\wedge$ trash entry with that origin	T1
permanent delete	path absent	T2
launch app	pid alive with creation time $t_c \wedge$ window title matches (pid, t_c) gone within timeout	T1/T3
terminate	foreground title matches pattern	T1/T2
focus window	accessibility value of element equals input	T0
set value	accessibility value of element equals input	T1/T3
toggle control	element state contains <i>checked</i>	T1/T2

cannot change after consent. Each local action is appended to a hash-chained audit log that stores digests of arguments and state, never file contents or typed text.

V. IMPLEMENTATION

The backend is Python (10,755 lines, 1,682 of them in the graph nodes) on FastAPI, LangGraph, Playwright and Pydantic, with SQLite for tasks and events and a persistent local Chroma store for vectors. The client is React in a Tauri shell; the Observatory is a separate FastAPI service with a React dashboard. The code is public at <https://github.com/sudeshsudhii/Agentic-pilot-local>. **[TODO: AUTHOR: confirm license; archive the evaluated version (tagged release with DOI).]**

a) Design decisions: The runner records each node’s name and status in a trace and checkpoints the state after every node, so a task resumed after a pause, an approval or a CAPTCHA re-enters the graph with its saved plan and skips intent parsing. The completion decision lives in one node; the terminal node adds a guard that downgrades a completion to FAILED if the intent named a site but navigation was never verified. Planner output always passes through deterministic rules, which corrects small models on known patterns but places part of the behavior outside the model.

b) Evidence: Per task, the evidence directory holds before/after screenshots of each executed action, screenshots after navigation and on verified completion, the latest DOM manifest, append-only verification and recovery records, a per-action execution record (action, URLs, screenshots, result, model, role, duration) and the node trace. A retry at the same model-call count overwrites the earlier screenshot pair, and files carry no integrity protection.

c) Status audit: Agent documentation tends to run ahead of the code, so we traced each mechanism to the path that

TABLE III
STATUS OF EACH MECHANISM AT COMMIT 2361d50, TRACED THROUGH THE CODE. I+T: IMPLEMENTED AND UNIT-TESTED; I: IMPLEMENTED, UNTESTED; P: PROPOSED OR PARTIAL. LIVE: AFFECTS THE DEFAULT EXECUTION PATH.

Mechanism	Status	Live	Note
Graph, checkpoints, resume	I+T	yes	compiled graph not tested
Completion predicate Φ	I+T	yes	rule-based (Sec. VII)
Per-step predicates, click check	P	–	steps advance on dispatch
CAPTCHA halt, resume, fallback	I+T	yes	no solver
DOM grounding, vision fallback	I+T	yes	vision on <code>need_help</code> only
Failure classes, retry bounds	I+T	yes	7 types, 6 attempts
Strategy-specific recovery	P	no	strategy label dropped
Role routing, stickiness	I+T	yes	cool-down not read
Memory write / retrieval	I / I+T	yes / no	retrieval bypassed
Sanitizer on element text	I+T	yes	title, URL unfiltered
Task-level approval gate	I	yes	model-assigned risk
Model-call audit log	I+T	yes	browsing not logged
Evidence store	I+T	yes	some files overwritten
Ablation flags	I	partial	verification flag unread
Desktop executor / extension	I / P	no	not wired (Sec. IV-G)

executes it (Table III). Most are live. Four are implemented but inert in a normal run: recovery strategies other than re-planning, memory retrieval, the routing cool-down and the verification ablation switch. The repository’s evaluation command calls scripts that are not in the repository, its ablation runner imports a function that does not exist, and its certification report counts a task as passed when it ends as completed and any screenshot and trace file exist. We use none of these repository-reported figures as results.

VI. EXPERIMENTAL SETUP

Component studies of the gate and the router need neither a language model nor network access and were run for this paper; end-to-end runs need local models and live sites, and we specify their protocol.

A. Component Studies

a) Environment: Linux container, 4-core Intel Xeon at 2.8GHz, 15 GiB RAM, no GPU; Python 3.11.15, Playwright 1.56.0 with headless Chromium 141.0.7390.37, LangGraph 1.2.12; repository commit 2361d50. Both study scripts are in the repository under `artifacts/experiments/`.

b) Gate study: We built 35 terminal browser states in the six categories of Table IV. Each consists of a goal, a URL, an HTML document served for that URL by Playwright request interception (no network traffic) and, where relevant, the extracted answer and the plan with step statuses, together with an oracle label Φ^* and a one-line rationale. Every case ends with a dispatched action, so a dispatch-only agent reports all of them complete. The gate under test is the agent’s own code: the `verify` node’s CAPTCHA pre-check followed by `verify_task_completion`. Decisions are deterministic;

latency is the median of 20 repetitions after a warm-up call. The authors wrote the cases and labels, including deliberately hard ones, so the rates describe the gate’s coverage, not its expected performance.

c) Routing study: We marked a set of models as installed in the registry and called the router with the arguments used at each call site in the agent, both in isolation and in the agent’s call order with the previous selection passed as the active model.

B. End-to-End Protocol

a) Hardware and models: Record CPU, GPU and VRAM, RAM, operating system, Ollama version and the digest of every model used. The default configuration uses `qwen2.5:1.5b` for text and `moondream` for vision; a second installs all configured models (Table V). The gateway currently sends no sampling options to Ollama, so each model’s default temperature applies; for reproducibility we set temperature to 0 and a fixed seed through the request options (a small code change) and run three trials per task. **[TODO: MEASURE: hardware, OS, Ollama version and model digests.]**

b) Task suite: The benchmark list shipped in the repository has eight tasks, four of which lack a checkable success criterion. We therefore propose 30 tasks in six categories of five: navigation, search, text entry into a form, extraction of a fact with a known answer, two- to three-step tasks, and blocking or error conditions (a public CAPTCHA demonstration page, HTTP error endpoints). Each task is paired with an oracle Φ^* written before any run, evaluated on the retained browser page or on the saved final screenshot and DOM snapshot by a person who does not see the agent’s reported status. **[TODO: AUTHOR: finalize the 30 tasks and their oracles; publish them with the code.]**

c) Configurations: (i) Full system. (ii) No completion gate: Φ replaced by “accept”, which requires making the `verify` node honor the existing but unused `enable_verification` flag. (iii) No recovery: `enable_recovery=false`, supported as is. (iv) Vision-only grounding: the DOM manifest withheld so that every action goes through the vision model, a small change. (v) Single model: supported as is. A memory ablation is meaningful only after the retrieval path is repaired (Section IV-F); until then, (vi) no memory is identical to (i).

d) Metrics: Task success rate under Φ^* ; FCR (Eq. 3); mean number of model calls and executed actions; mean wall-clock time per task; recovery success rate (tasks that entered recovery and ended satisfied under Φ^*); vision-invocation rate (fraction of actions grounded by the vision model); and evidence completeness (fraction of executed actions with both screenshots, an execution record and, for completed tasks, a completion-proof screenshot). Rates are reported with Wilson 95% intervals over task–trial pairs.

TABLE IV

COMPLETION DECISIONS ON 35 CONSTRUCTED END STATES. n^+/n^- : CASES WHOSE GOAL IS / IS NOT SATISFIED BY THE ORACLE. REP.: CASES REPORTED COMPLETE. FC: FALSE COMPLETIONS (REPORTED COMPLETE, GOAL NOT SATISFIED). FR: FALSE REJECTIONS (GOAL SATISFIED, NOT REPORTED COMPLETE). DISPATCH-ONLY REPORTS EVERY CASE COMPLETE. SOURCE: VERIFICATION_GATE_SUMMARY.CSV.

Category	n^+	n^-	Dispatch-only		Evidence gate		
			Rep.	FC	Rep.	FC	FR
Search	3	4	7	4	4	1	0
Navigation	2	5	7	5	4	2	0
CAPTCHA	0	4	4	4	1	1	0
Text entry	3	3	6	3	4	1	0
Extraction	3	4	7	4	3	1	1
Multi-step	2	2	4	2	3	1	0
All	13	22	35	22	19	7	1
FCR				0.629		0.368	

VII. RESULTS AND DISCUSSION

A. Completion Gate on Constructed End States

Table IV reports the gate study. Of the 22 end states that do not satisfy their goal, the gate withholds completion in 15: all three search-engine home pages, all three pages with an HTTP error code in the title, three of the four bot-check pages (halted as BLOCKED), an empty and a wrong text field, three incomplete extraction answers including one produced while still on the search home page, and a plan with a pending step. The dispatch-only rule reports all 22 as complete. Among the 13 satisfied end states, the gate rejects one.

The FCR values depend on the case mix we chose and are not expected rates on real tasks; the case-level outcomes are the informative part. Each of the seven false completions traces to a specific gap in Φ . A results page for a different query passes, because the search check only asks whether the URL is a results page (S7). A “page not found” message without a status code in the title and a redirect to a login page pass (N6, N7); the DOM extractor classifies such pages, but Φ does not consult that classification. An interstitial containing none of the 12 known challenge phrases passes (C4). “Meeting at 10am tomorrow” satisfies a request to enter exactly “Meeting at 10am”, because matching is by substring (T4). Any answer over 40 characters that does not begin with a navigation verb counts as an extraction even without the requested fact (X4); since the planner’s own reasoning string on a complete action becomes the answer, a verbose planner can satisfy this check by itself. Finally, a click dispatched on the wrong element marks its step complete, so the plan check passes (P3). The one false rejection (X6) mirrors X4: a correct 13-character answer fails the length test. Every gap maps to a check that needs no model call (query comparison, reuse of the page-state classifier, exact matching, answer-to-goal term overlap, step predicates from the planner’s expected outcome), and the fixtures can serve as a regression suite for them.

The gate is cheap relative to model inference. On a 4-core Intel Xeon (2.8GHz) container with Chromium 141 in headless mode, the median time for one full decision

TABLE V

MODEL SELECTED BY THE ROUTER PER CALL TYPE (NO INFERENCE PERFORMED). DEFAULT: QWEN2.5:1.5B AND MOONDRREAM INSTALLED.

FULL: ALL CONFIGURED MODELS INSTALLED (QWEN2.5:1.5B, QWEN2.5:7B, QWEN3.5:2B, DEEPSEEK-R1:1.5B, QWEN2.5-CODER:3B, QWEN3-VL:2B, MOONDRREAM). ISOLATED: EACH CALL ROUTED WITH NO ACTIVE MODEL. AGENT ORDER: CALLS ROUTED IN GRAPH ORDER WITH THE PREVIOUS SELECTION AS THE ACTIVE MODEL. ^fSELECTED BY THE FALLBACK CHAIN. SOURCE: ROUTING_RESOLUTION.CSV.

Call (role)	Default	Full, isolated	Full, agent order
Intent parsing (light.)	qwen2.5:1.5b	deepseek-r1:1.5b	deepseek-r1:1.5b
Decompose (planner)	qwen2.5:1.5b ^f	qwen3.5:2b	deepseek-r1:1.5b
1-step act. (light.)	qwen2.5:1.5b	deepseek-r1:1.5b	deepseek-r1:1.5b
Multi-step act. (planner)	qwen2.5:1.5b ^f	qwen3.5:2b	deepseek-r1:1.5b
Recovery (recovery)	qwen2.5:1.5b ^f	qwen2.5:7b	qwen2.5:7b
Image input (vision)	moondream	qwen3-vl:2b	qwen3-vl:2b

(CAPTCHA pre-check plus Φ) over 20 repetitions per case was 33.8ms, with a maximum per-case median of 52.5ms; decisions that stop at the URL-based CAPTCHA check took under 8ms. Most of the time is spent reading page content and title through the browser protocol.

B. Routing Behavior

Table V shows the router’s selection per call type, computed without inference. With the launcher’s default installation, every text call resolves to qwen2.5:1.5b, three of them through the fallback chain, so routing reduces to one text and one vision model. With all configured models installed, isolated calls each receive their role’s preferred model. In the agent’s actual call order, however, stickiness keeps the lightweight model chosen for intent parsing (deepseek-r1:1.5b, whose profile lists planning) for decomposition and multi-step planning; only recovery, which is exempt, and image inputs switch. Stickiness saves model loads at the cost of never reaching the planner model, a trade the single-model ablation can quantify.

C. Test Suite

Table VI summarizes the repository’s tests. All passing tests run without a language model: node tests call graph nodes directly with the executor and gateway stubbed, no test runs the compiled graph, and the two passing end-to-end tests accept a failed task as well as a completed one. Of the four failures, two need Ollama, one times out waiting on a browser page and one needs scripts missing from the repository; the collection error is a missing import under Python 3.11. The suite supports claims about component behavior (CAPTCHA blocking, routing rules, checkpointing), not about task success.

D. End-to-End Evaluation

End-to-end results require a local Ollama installation and live websites, neither of which was available in the environment used to prepare this draft. **[TODO: MEASURE: run the protocol of Section VI and report, per configuration, task success rate, FCR under the independent oracle, mean model calls, mean wall-clock time**

TABLE VI
REPOSITORY TEST SUITE IN THE COMPONENT-STUDY ENVIRONMENT,
WITHOUT OLLAMA (HEADLESS CHROMIUM, 240 S PER-TEST TIMEOUT).
ERR.: COLLECTION ERROR. LINE COVERAGE OF BACKEND/: 62.0 %.
SOURCE: TESTS_BY_FILE.CSV, JUNIT.XML.

Test group (files)	Tests	Pass	Fail	Err.
Agent nodes, CAPTCHA, checkpoint, recovery (6)	35	33	2	1
Model router, registry, parser (4)	20	20	0	0
RAG and memory (8)	28	28	0	0
Config, evidence, security, telemetry (1)	11	11	0	0
Database, plugins, launcher (3)	11	10	1	0
End-to-end with browser and LLM (2)	3	2	1	0
Total, tests/ (24)	108	104	4	1
Observatory (1)	20	20	0	0

per task, recovery success rate, vision-invocation rate and evidence completeness, with 95% confidence intervals over trials.] [TODO: MEASURE: add a qualitative case study with before/after screenshots from the evidence directory (~/.pilot/logs/evidence/<task_id>/) of one task in which the gate rejected a premature completion and the agent then finished the task.]

VIII. SECURITY, PRIVACY AND ETHICAL CONSIDERATIONS

a) Threat model: We consider (i) instructions embedded in page content that try to redirect the agent [27]; (ii) high-impact actions such as posting, sending or purchasing without consent; (iii) damage to the local system once file and process actions exist; and (iv) disclosure of user data to unintended parties. The user and the host are trusted; websites are not. A compromised host or model is out of scope.

b) Current mitigations: A sanitizer replaces 17 patterns of instruction-override, delimiter, exfiltration and role-manipulation phrasing in element text, labels and placeholders before they reach the planner; retrieved knowledge is filtered the same way and marked untrusted. Before navigation, a risk gate pauses tasks whose action is posting, e-mailing, purchasing, deleting or transferring, or whose risk level is high or critical, and asks the user; the check is repeated when completion is verified. The service listens on loopback and accepts only localhost origins. Every model call is logged with destination, a locality flag and payload sizes, and credential-like values are scrubbed from broadcast events.

c) Gaps: The sanitizer is a fixed denylist; systematic evaluations find existing prompt-injection defenses insufficient [28], and paraphrased or encoded instructions do not match its patterns. The page title, URL, extracted text and screenshots are not filtered. The risk level that triggers approval comes from the same model that reads untrusted content, and one approval covers every later action of the task. The audit log covers model calls but not browsing, and a non-local model address is logged, not blocked. The privacy property we claim is therefore narrow: *by default, model inference runs on the user's machine and no prompt or screenshot is sent to a model provider.* Data the user asks

the agent to enter on a website still leaves the machine. The desktop design closes some of these gaps with deterministic tiers, action-bound approvals and a hash-chained audit log.

d) CAPTCHAs and automation etiquette: On a CAPTCHA or bot check the agent stops, marks the task BLOCKED and keeps the browser open; the user can solve the challenge and resume (refused while it is still present) or restart on another search engine. There is no solver. The browser does, however, disable automation-detection features, send a desktop user-agent string and hide navigator.webdriver, and it types queries instead of loading result URLs. This suits a personal assistant acting at human pace on its user's behalf, not bulk use, and deployments should respect site terms. Chromium also runs with its sandbox disabled; re-enabling it is a simple hardening step.

IX. LIMITATIONS AND FUTURE WORK

a) Evidence: Only the completion gate and the router are measured here, and the gate only on constructed end states. End-to-end rates and latencies with local models remain to be measured (Section VI); latency will depend strongly on the host and on which models are installed.

b) Rule-based verification: Φ is keyed to request wording and a few search engines, and Section VII lists its blind spots. We plan to derive typed step predicates from the planner's expected outcomes, add a local vision-model check of the final screenshot as a second opinion, and calibrate both on labelled trajectories.

c) Small models and hand-written rules: Planning with 1.5B–7B models is weak, and deterministic overrides compensate on common patterns, some of them site-specific. The extraction handler also contains post-processing written for one case-study site. **[TODO: AUTHOR: remove or generalize the case-study-specific extraction code in backend/agent/nodes.py (lines 1108–1139) before any end-to-end evaluation, or disclose it here.]** Canvas- and WebGL-rendered pages expose no useful DOM and depend on the vision path.

d) Inactive mechanisms and roadmap: Strategy-specific recovery, memory retrieval, the routing cool-down and the verification switch need small fixes to take effect (Table III); the ablations are meant to measure them once they do. For the local-system extension we will implement the path and process policy, the predicate library, an AT-SPI provider (testable headlessly on Linux) and the `action_gate` node, then Windows UI Automation. Longer term, we plan to split planning, grounding and verification across cooperating agents, with the verifier given no write access to the environment.

X. CONCLUSION

Agentic Pilot runs a browser agent entirely against local models and refuses to report success until a check on the observed page passes, keeping the observations behind each decision on disk. Its architecture separates observation, planning, execution, verification and recovery into explicit graph

nodes, which made it possible to state the completion rule formally and to test it in isolation. On constructed end states the rule removes most of the premature completions a dispatch-only agent would report, and its failures point to specific checks that are missing. The same structure carries over to files, processes and native applications once accessibility trees, typed postconditions and deterministic permission tiers take the place of the DOM, page checks and model-assigned risk. The remaining work is empirical: measuring the full system with local models on real tasks, and activating the recovery and memory paths that are designed but not yet live.

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